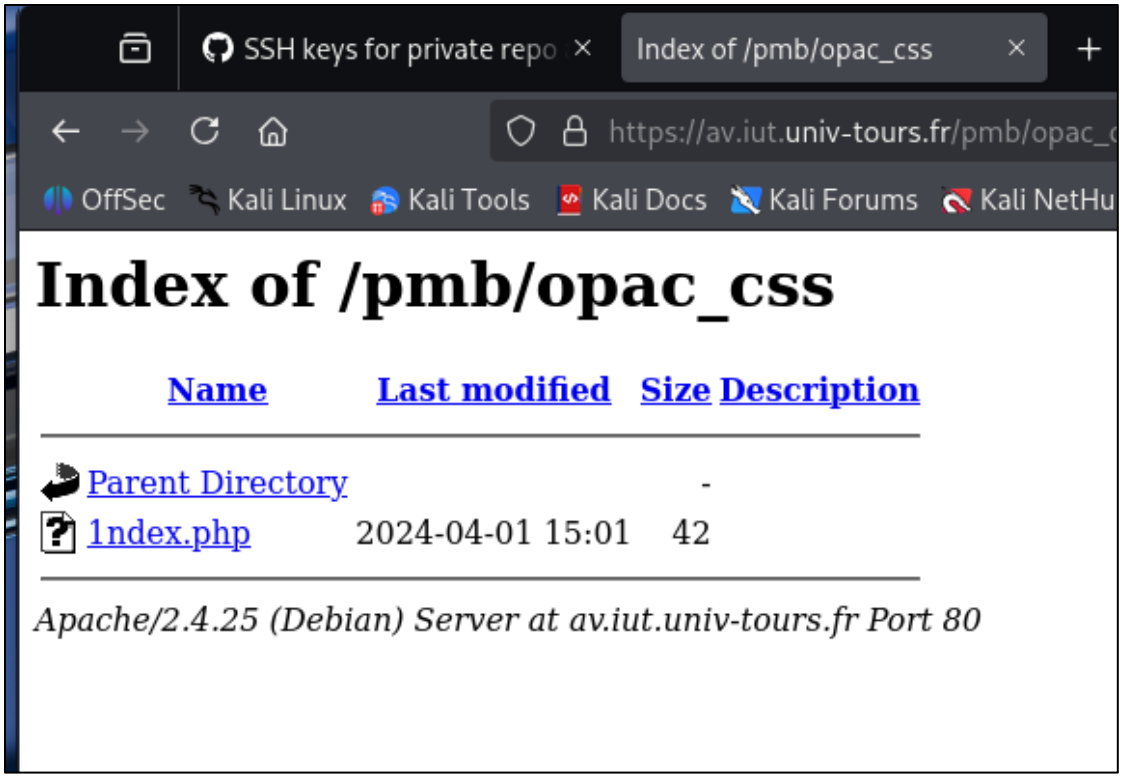
Name: Redenelle Maurice L. Manangu **Section: CS-302**

Midterm Lab Activity 1: Using Exploit Database (Google Hacking Database)



"PMB" AND ("changelog.txt" OR inurl:opac_css)

1. **The purpose or objective of the dork.**

- This dork searches for PMB installations, an open-source library management system, by identifying sites containing "PMB" and either changelog.txt files or URLs containing opac css (Online Public Access Catalog CSS files).

2. **To what extent can a hacker exploit the dork.**

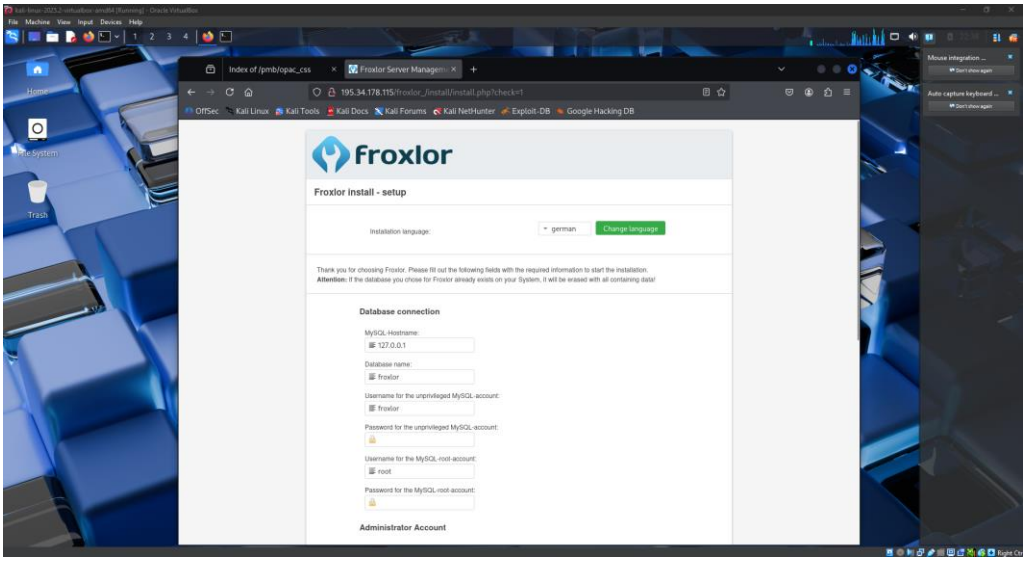
- The process involves identifying publicly accessible library management systems, locating changelog.txt files, identifying unprotected admin interfaces, and discovering databases containing library records and administrative credentials. Targeting known vulnerabilities specific to identified PMB versions, accessing sensitive library patron data, and identifying specific exploitation vectors such as default credentials, SQL injection vulnerabilities, directory traversal attacks, and cross-site scripting in public catalog interfaces.

3. **What steps can the resource's owner take to protect it?**

- To protect library management systems from attacks, it's crucial to implement access control measures, system hardening, regular security audits, and proper backup and recovery procedures. These measures include restricting access to administrative directories and sensitive files, ensuring the latest version of PMB software is used, protecting development/debug files, and implementing proper file permissions on the web server. Regular security audits and backup procedures can help identify and target library management systems.

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inurl:install.php intitle:"Froxlor Server Management Panel - Installation"

1. **The purpose or objective of the dork.**

- This dork identifies exposed Froxlor installation pages by using two search operators: inurl:install.php, targeting URLs with installation script, and intitle:"Froxlor Server Management Panel - Installation," matching exact page titles, thereby enabling a highly targeted search for incomplete Froxlor installations.

2. **To what extent can a hacker exploit the dork.**

- Critical Exploitation Opportunities involve completing the installation process with attacker-controlled credentials, harvesting database credentials, creating administrative accounts with full server management privileges, accessing the MySQL root account, controlling the entire hosting infrastructure, deploying web shells or backdoors, modifying DNS settings, and accessing customer data and hosting account information. The attack chain includes finding exposed installations via the dork, completing the installation with malicious admin credentials, gaining access to the Froxlor administrative panel, exploiting hosting control features, and compromising all hosted websites and services.

3. **What steps can the resource's owner take to protect it?**

- To ensure web application security, delete all installation files immediately after installation, rename them during setup, and use.htaccess restrictions to limit access to specific IP addresses. Implement IP whitelisting during the installation phase, use HTTP authentication, change default database names and usernames, and generate strong passwords for all accounts. Post-installation, verify the removal of all installation-related files, change default administrative paths, enable two-factor authentication, implement SSL/TLS for administrative interfaces, and conduct regular security audits. Monitor access logs for attempts to access installation URLs and keep Froxlor updated with the latest security patches.